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Simulating random variables using moment generating functions and the saddlepoint approximation

Don McLeish
Department of Statistics & Actuarial Science
University of Waterloo

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Don McLeish
Department of Statistics and Actuarial Science, University of Waterloo

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Abstract

When we are given only a transform such as the Laplace transform, moment generating function or characteristic function of a distribution, it is rare that we can simulation efficiently simulate random variables. Possible approaches such as inverse transform using numerical inversion of the transform are computationally very expensive. However, the saddlepoint approximation is known to be exact for many distributions including the Normal, Gamma, and inverse Gaussian and remarkably accurate for a large number of others. We explore the efficient use of the saddlepoint approximation for simulating distributions and provide three examples of the accuracy of these simulations.

Key-words: simulation, saddlepoint approximation, sum of gamma random variables, stochastic volatility, Heston model, Feller process, sum of uniform random variables

1 Introduction

There are many techniques for simulating random variables based on either the cumulative distribution function (inverse transform) or the probability density function (for example acceptance-rejection), but there are also cases where neither of these methods can be applied becasue the information on the distribution is provided in a less tractible form. We may, for example, be given only a transform of the density such as the Laplace transform, moment generating function or characteristic function. In these cases, it is rare that we can provide an exact simulation from the distribution. One can attempt to invert the transform numerically to obtain the c.d.f. for example, and then invert this, using inverse transform methods. However this is compuationally very intensive and therefore virtually impossible when one wishes to do a large number of simulations. An easy alternative is to use a saddlepoint approximation to the density which is well-known to be exact for families of distributions including the Normal, the

Gamma and the inverse Gaussian distribution, but, in general, is quite accurate whenever a distribution is obtainable as a sum or a mean of independent random variables. See Barndorff-Neilson and Cox (1989) for a discussion of saddlepoint approximations and their generalisations.

It is the purpose of this note to explore the use of the saddlepoint approximation for simulating distributions that are otherwise intractable or computationally demanding.

2 Simulating from the Saddlepoint Approximation

Suppose we wish to generate a random variable having known cumulant generating function $k(t) = \ln E(e^{tX})$. The Lugannani and Rice(1980) saddlepoint approximation is

$$P(X \geq x) \simeq 1 - \Phi(w) + \phi(w)\left(\frac{1}{u} - \frac{1}{w}\right) \tag{1}$$

where

$$\begin{aligned} w &= w(t) = \sqrt{2(tk'(t) - k(t)}sgn(t) \\ u &= u(t) = t\sqrt{k''(t)} \end{aligned}$$

and t solves $k'(t) = x$. Let $g(x)$ be the inverse function of k' so that the solution above is $t = g(x)$. Consider the distribution of the random variable $g(X)$ under the approximation (1). This implies, with $t = g(x)$,

$$\begin{aligned} P(g(X) &\leq t) \simeq \Phi(w(t)) + \phi(w(t))\left(\frac{1}{w(t)} - \frac{1}{u(t)}\right) \qquad (2) \\ w(t) &= \sqrt{2(tk'(t) - k(t)}sgn(t) \\ u(t) &= t\sqrt{k"(t)} \end{aligned}$$

Suppose we generate a random variable T having cdf $\Phi(w(t))+\phi(w(t))\left(\frac{1}{w(t)} - \frac{1}{u(t)}\right)$. Then the random variable $X = k'(T)$ has cdf given by the saddlepoint approximation (1). The density corresponding to the cdf (2) is (see Barndorff-Neilson and Cox (1989) section 4.50, page 110) to first order

$$\sqrt{k''(t)}\phi(w(t)) = \sqrt{\frac{k''(t)}{2\pi}}\exp(k(t) - tk'(t)) \tag{3}$$

This is the saddlepoint approximation to the density function of the distribution of the maximum likelihood estimator in an exponential family constructed by tilting the original measure. Note that its mode m satisfies $k'''(m) = 2t\left(k''(m)\right)^2$ and this is similar to the probability density function we would obtain if we assumed that the random variable $w(T)$ had a standard normal

distribution. In this latter case, assuming the function $w(t)$ is monotone, the pdf is

$$\phi(w(t))\left|\frac{dw}{dt}\right| = \phi(w(t))\left|\frac{tk''(t)}{w(t)}\right|$$

and these densities are similar if $w(t) = \sqrt{\frac{tk'(t)-k(t)}{k''(t)}}$ is approximately proportional to $|t|$. In order to generate random variables from a density proportional to (3) we have several alternatives.

a. We can use **inverse transform** directly from the cdf. i.e. with $F(t) = \Phi(w(t)) + \phi(w(t))\left(\frac{1}{w(t)} - \frac{1}{u(t)}\right)$. we generate a uniform$[0,1]$ random variable U, and then solve $U = F(t)$ using, for example, a Newton-Raphson iteration

$$t_{n+1} = t_n - \frac{F(t_n) - U}{\sqrt{k''(t_n)}\phi(w(t_n))}$$

This is relatively slow since it requires obtaining a root for each variable generated.

b. We can use **acceptance rejection**. As long as we can dominate the density $\sqrt{k''(t)}\phi(w(t))$ with a multiple of a simple probability density function, this is a reasonably efficient option. For example if the function $\ln\left(\sqrt{k''(t)}\phi(w(t))\right) = -tk'(t) + k(t) + \frac{1}{2}\ln(k''(t))$ is a concave function for $t > m$, then (see Devroye (1986), p.289-291) there is a simple rejection algorithm for generating from this distribution. In other particular cases, such as those below, we can seek an alternative deominating function.

c. When it is difficult to dominate (3), we can use a **Metropolis-Hastings MCMC** algorithm, providing a Markov process with the correct limiting distribution. A choice of proposal density may be motivated by the fact that the saddlepoint density (3) is close to the density

$$k''(t)\exp(k(t) - tk'(t))$$

obtained by assuming that $tk'(t) - k(t)$ has a standard exponential distribution or more simply use a local linear approximation to the log density $-tk'(t) + k(t) + \frac{1}{2}\ln(k''(t))$, equivalent to a proposal density which is an exponential distribution conditioned to a neighbourhood of the current value.

In the examples below we use choice (b) since it is the simplest, often the least computationally expensive, and provides for pseudo-independent observations.

Example 1 *Sum of Gamma random variables*

Many distributions are expressible as a sum of random variables each of which has a gamma distribution. This is the case, for example, with the Anderson-Darling Statistic (see for example Murakami (2009)). Suppose we

wish to simulate values of the random variables $X = \sum_i Y_i$ where the random variables Y_i are independent $Gamma(\alpha_i, \beta_i)$. This may be a finite or infinite sum, but in the latter case, obviously we require rates of convergence of the parameters, for example we may require $\sum_i \alpha_i \beta_i < \infty$ and $\sum_i \alpha_i \beta_i^2 < \infty$. The cumulant generating function and its derivatives are given below.

$$\begin{aligned}
k(t) &= -\sum_i \alpha_i \ln(1-\beta_i t) \text{ for } t < \min(\frac{1}{\beta_i}) \\
k'(t) &= \sum_i \frac{\alpha_i \beta_i}{1-\beta_i t} \\
k''(t) &= \sum_i \frac{\alpha_i \beta_i^2}{(1-\beta_i t)^2} \\
k'''(t) &= \sum_i \frac{2\alpha_i \beta_i^3}{(1-\beta_i t)^3} \\
tk'(t) - k(t) &= \sum_i \alpha_i \left[\frac{\beta_i t}{1-\beta_i t} + \ln(1-\beta_i t) \right]
\end{aligned}$$

The saddlepoint density becomes

$$\sqrt{\frac{k''(t)}{2\pi}} \exp(k(t) - tk'(t)) = \sqrt{\frac{k''(t)}{2\pi}} \exp(-\sum_i \alpha_i \frac{\beta_i t}{1-\beta_i t}) \prod_i (1-\beta_i t)^{-\alpha_i} \text{ for } t < \min(\frac{1}{\beta_i})$$

Consider the behaviour of the density at its two extremes, as $t \to -\infty$ and $t \to \min(1/\beta_i)$. As $t \to -\infty$, $k''(t) \to 0$, $\exp(-\sum_i \alpha_i \frac{\beta_i t}{1-\beta_i t}) \to \exp(\sum_i \alpha_i)$ and $\prod_i (1-\beta_i t)^{-\alpha_i} \sim |t|^{-\sum_i \alpha_i}$. Therefore $\sqrt{\frac{k''(t)}{2\pi}} \exp(k(t) - tk'(t)) = o(|t|^{-\sum_i \alpha_i})$. As $t \to 1/\beta_i$, consider the term

$$\begin{aligned}
\sqrt{\frac{1}{(1-\beta_i t)^2}} \exp(-\alpha_i \frac{\beta_i t}{1-\beta_i t})(1-\beta_i t)^{-\alpha_i} &= \exp(-\alpha_i \frac{\beta_i t}{1-\beta_i t})(1-\beta_i t)^{-1-\alpha_i} \\
&= e^{\alpha_i} \exp(-\alpha_i \frac{1}{1-\beta_i t})(1-\beta_i t)^{-1-\alpha_i} \to 0
\end{aligned}$$

Therefore we can always bound the saddlepoint density with a function of the form

$$c \min(1, (1-bt)^{-a}) \tag{4}$$

for $1 < a < \sum_i \alpha_i$ and suitable constants $b = \max(\beta_i)$, c. It is easy to generate from a density proportional to (4), namely $\frac{b(a-1)}{a} \min(1, (1-bt)^{-a})$ .by inverse transform since the corresponding cdf and its inverse are:

$$\begin{aligned}
F(t) &= \begin{cases} \frac{1}{a}(1-bt)^{-a+1}, & t < 0 \\ \frac{1}{a} + (1-\frac{1}{a})bt, & 0 < t < \frac{1}{b} \end{cases} \\
F^{-1}(U) &= \begin{cases} \frac{1}{b}\left(1 - (aU)^{-1/(a-1)}\right), & 0 < U < a^{-1} \\ \frac{aU-1}{b(a-1)}, & a^{-1} \le U < 1 \end{cases}
\end{aligned} \tag{5}$$

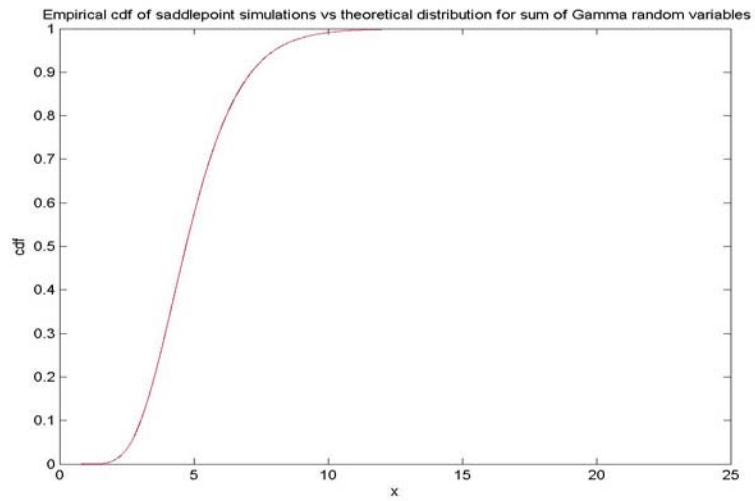

Figure 1: Empirical cdf of saddlepoint simulations vs theoretical distribution for sum of Gamma random variables

Having found the parameters a, c in (4), the acceptance rejection algorithm is as follows:

1. Generate $U \sim U[0,1]$ and define $T = F^{-1}(U)$ using (5).
2. Generate $V \sim U[0,1]$.
3. If $V \leq \frac{\sqrt{\frac{k''(T)}{2\pi}} \exp(k(T) - Tk'(T))}{c \min(1,(1-bT)^{-a})}$ output the value $k'(T)$.Otherwise return to step 1.

We chose as an example a sum of 20 exponential random variables with $\alpha_i = 1, \beta_i = (0.8)^{i-1}$, for $i = 1, 2, ..., 20$. The scale parameters differ by a factor of up to almost 70. In a simulation of 1,000,000 runs of the AR algorithm resulted in 565,677 draws from the saddlepoint density. The empirical c.d.f. is plotted in Figure (1), together with the empirical cdf of values simulated from the correct distribution. Clearly these two plots are essentially identical, with very small differences in the remote tails ($x > 13$) where both cdfs are greater than 0.999. The mean and variance of the simulated random variables agreed well with the theoretical counterparts, $\sum_i \alpha_i \beta_i$ and $\sum_i \alpha_i \beta_i^2$.

Example 2 *Simulating a Volatility Process*

A Feller or CIR process is a stochastic process V_t driven by the stochastic differential equation

$$dV_t = \kappa(\theta - V_t)dt + \sigma_v \sqrt{V_t} dW(t), V_0 = v_0 > 0 \tag{6}$$

where W is a Brownian motion processes. In the Heston stochastic volatility model, V_t is the volatility process and it is of particular interest to simulate jointly the spot volatility V_T at the end of a period and the aggregate volatility over this period $\int_0^T V_s ds$. The marginal distribution of V_T is well-known (see Glasserman) and Glasserman and Kim (2009) obtain the Laplace transform of the conditional distribution of $\int_0^T V_s ds$ given V_0 and V_T. This is described as an integral over a Bessel distributed random variable η so that, translated to our notation, this results in a moment generating function

$$E\left[e^{t\int_0^T V_s ds}|V_0, V_T\right] = E\left[\exp\{\omega\left[\kappa\coth\kappa\gamma - L\coth L(t)\gamma\right]\} + \left(\frac{\delta}{2} + 2\eta\right)\left(\frac{L\sinh\kappa\gamma}{\kappa\sinh L\gamma}\right)\right]$$

where the outer expectation on the right side is over the distribution of a Bessel random variable η having parameters $\nu = \frac{\delta}{2} - 1$ and $z = \frac{2\kappa/\sigma^2}{\sinh(\kappa\gamma)}\sqrt{V_0 V_T}$ and where

$$\begin{aligned}
\delta &= \frac{4\kappa\theta}{\sigma^2} \\
L &= L(t) = \sqrt{\kappa^2 - 2\sigma^2 t}, \text{ and } L' = -\frac{\sigma^2}{L} \\
\gamma &= \frac{T}{2}, \\
\omega &= \frac{v_0 + v_T}{\sigma^2}.
\end{aligned}$$

With some additional notation

$$\begin{aligned}
c &= c(t) = \coth L(t)\gamma, \text{ and } c' = -(1 - c^2)\frac{\sigma^2\gamma}{L} \\
\varkappa &= \frac{\delta}{2} + 2\eta,
\end{aligned}$$

We can write the cumulant generating function of the distribution conditional on V_0, V_T, η as:

$$k(t) = \omega\left[\kappa\coth\kappa\gamma - Lc\right] + \varkappa\left[\ln(L\sinh\kappa\gamma) - \ln(\kappa\sinh L\gamma)\right]$$

Our objective is to simulate efficiently from this distribution. The derivatives are:

$$
\begin{aligned}
k'(t) &= \omega\left[-L'c - Lc'\right] + \varkappa\left[\frac{L'}{L} - \frac{\sinh' L\gamma}{\sinh L\gamma}L'\gamma\right] \\
&= \frac{\sigma^2\omega}{L}\left[c + (1-c^2)L\gamma\right] + \frac{\sigma^2}{L^2}\varkappa\left[-1 + L\gamma\frac{\cosh L\gamma}{\sinh L\gamma}\right] \\
&= \frac{\sigma^2\omega}{L}\left[c + (1-c^2)L\gamma\right] - \frac{\sigma^2}{L^2}\varkappa\left[1 - L\gamma c\right] \\
k'(0) &= \frac{\sigma^2\omega}{\kappa}\left[\coth\kappa\gamma + (1-\coth^2\kappa\gamma)\kappa\gamma\right] - \frac{\sigma^2}{\kappa^2}\varkappa\left[1 - \kappa\gamma\coth\kappa\gamma\right]
\end{aligned}
$$

$$
\begin{aligned}
k''(t) &= -\frac{\sigma^2\omega}{L^2}L'\left[c + (1-c^2)L\gamma\right] + \frac{\sigma^2\omega}{L}\left[c' + (1-c^2)L'\gamma - 2cc'L\gamma\right] \\
&\quad + 2\frac{\sigma^2}{L^3}L'\varkappa\left[1 - L\gamma c\right] + \frac{\sigma^2}{L^2}\varkappa\left[L'\gamma c + L\gamma c'\right] \\
\frac{k''(t)L}{\sigma^4} &= (\frac{\omega + \gamma\varkappa}{L})(\frac{c}{L} - (1-c^2)\gamma) + 2c\omega\gamma^2(1-c^2) - 2\frac{\varkappa}{L^3}
\end{aligned}
$$

We will try dominating the saddlepoint density $\sqrt{\frac{k''(T)}{2\pi}}\exp(k(T) - Tk'(T))$ with a multiple of a density function similar to that used for the sum of gamma distributed random variables, having cumulative distribution function $F(t) = \begin{cases} \frac{1}{a}(1-bt)^{-a+1}, & t < 0 \\ \frac{1}{a} + (1-\frac{1}{a})bt, & 0 < t < \frac{1}{b} \end{cases}$ where $a = 2, b = \frac{2\sigma^2}{\kappa^2}$. We used parameters as follows

$$\kappa = 6 \quad \sigma = 0.6 \quad \delta = 1 \quad \eta = 1 \quad T = 1 \quad V_0 = .5, V_T = 1$$

In this case, $\omega = \frac{v_0 + v_T}{\sigma^2} = \frac{1.5}{0.6^2} = 4.1667, \gamma = \frac{T}{2} = 0.5, \varkappa = \frac{\delta}{2} + 2\eta = 2.5$

$$k'(0) = \frac{0.6^2 \times 4.1667}{6}\left(\coth 3 + (1-\coth^2 3)3\right) - \frac{0.6^2}{6^2}2.5\left(1 - 3\coth 3\right) = 0.29414$$

so that the mean of a simulated sample should be around this value. We give the saddlepoint density for T and the dominating function in Figure (2). The constant $c = 0.023$ was used to dominate the saddlepoint density. We simulated 217,845 values from the above conditional distribution using acceptance-rejection and obtained a sample mean of .2943, which compares well with the theoretical value above, and a sample variance of 0.0024. This simulation required less than one second of CPU time on a PC running Matlab with 2 Intel CPU at 2 GZ. The probability histogram of the sample is displayed in Figure (**??**).

Example 3 *Sum of n Uniform[-1,1]*

Consider a random variable defined as

$$X = \sum_{i=1}^{n} Y_i$$

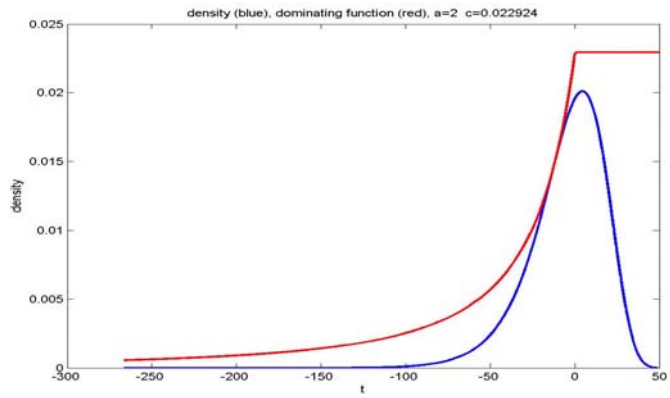

Figure 2: The saddlepoint density for T together with the dominating function $c\min(1,(1-bt)^{-a}), c = 0.023$

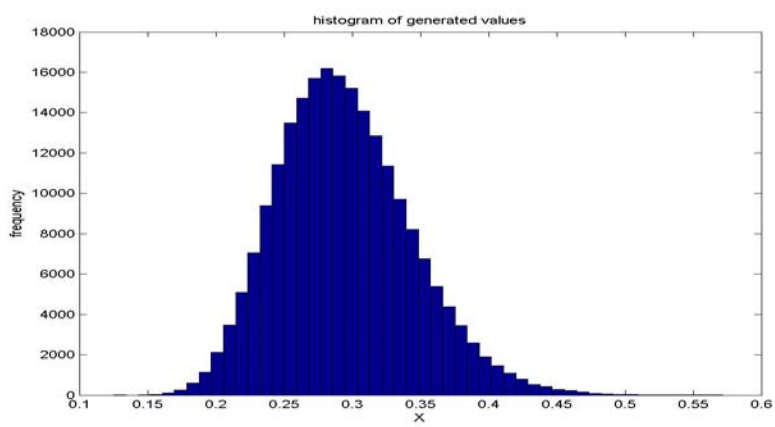

Figure 3: Histogram of generated sample of aggregate volatility

where each random variable Y_i is Uniform[-1,1]. The cumulant generating function of X and its derivatives are:

$$\begin{aligned}
k_n(t) &= n\ln\left(\frac{\sinh(t)}{t}\right) = n\ln(\frac{\sinh(t)}{t}) \\
k'(t) &= n\frac{d}{dt}\left(\ln(\sinh(t)) - \ln(t)\right) = n\left(\frac{1}{\tanh t} - \frac{1}{t}\right) \\
k''(t) &= n\frac{d}{dt}\left(\frac{\cosh t}{\sinh t} - \frac{1}{t}\right) = n\left(1 + \frac{1}{t^2} - \frac{1}{\tanh^2 t}\right) \\
-tk'(t) + k(t) + \frac{1}{2}\ln(k''(t)) &= n + \frac{1}{2}\ln(n) - nt\frac{\cosh t}{\sinh t} + n\ln(\frac{\sinh(t)}{t}) + \frac{1}{2}\ln\left(1 + \frac{1}{t^2} - \frac{\cosh^2 t}{\sinh^2 t}\right) \\
&= \text{constant} - n\left(\frac{t}{\tanh t} - \ln(\frac{\sinh(t)}{t}) - \frac{1}{2n}\ln\left(1 + \frac{1}{t^2} - \frac{1}{\tanh^2 t}\right)\right) \\
&\sim \text{constant} - (n+1)\ln(t) + o(\ln(t))
\end{aligned}$$

since $0 \le 1 + \frac{1}{t^2} - \frac{1}{\tanh^2 t} \le 1$ for all t and $\frac{1}{\tanh^2 t} = 1 + 4e^{-2t} + O(e^{-4t})$. This implies that the saddlepoint density is bounded above by a function with tails of the form $|t|^{-p}$ provided that $p < n+1$. For $n > 2$, we can choose the Cauchy distribution to dominate the density since for suitable constant c,

$$\sqrt{\frac{k_n''(t)}{2\pi}}\exp(k_n(t) - tk_n'(t)) \le c\frac{b}{\pi}\frac{1}{b^2 + t^2}$$

We chose the scale constant of the Cauchy $b = \sqrt{\frac{n}{4}}$ to provide a reasonable fit. The saddlepoint density is quite simple:

$$\sqrt{\frac{k_n''(t)}{2\pi}}\exp(k_n(t) - tk_n'(t)) = \sqrt{\frac{n}{2\pi}}e^n\sqrt{1 + \frac{1}{t^2} - \frac{1}{\tanh^2 t}}\left(\frac{\sinh(t)}{t}\right)^n\exp(-\frac{nt}{\tanh t})$$

The actual density function of $X = k'(T)$ is given by (see Devroye, p. 732) using the inversion of the characteristic function $\left(\frac{\sin(t)}{t}\right)^n$

$$\begin{aligned}
f(x) &= \frac{1}{\pi}\int_0^\infty \left(\frac{\sin(t)}{t}\right)^n \cos(tx)dt \\
&= \frac{1}{(i-1)!}\frac{1}{2}\sum_{k=0}^{i-1}(-1)^k\binom{i}{k}(x - (2k-n))^{i-1} \quad \text{for } 2i - 2 - n < x < 2i - n; i = 1, 2, ..., n
\end{aligned}$$

The acceptance-rejection algorithm is as follows:

- Generate $U \sim U[0,1]$. Generate $T = b/\tan(\pi U)$ from a Cauchy$(0, b)$ distribution with $b = \sqrt{\frac{n}{4}}$.
- Generate $V \sim U[0,1]$. If

$$V < \frac{\sqrt{n\pi}e^n}{cb\sqrt{2}}\sqrt{1 + \frac{1}{T^2} - \frac{1}{\tanh^2 T}}\left(\frac{\sinh(T)}{T}\right)^n\exp(-\frac{nT}{\tanh T})(b^2 + T^2)$$

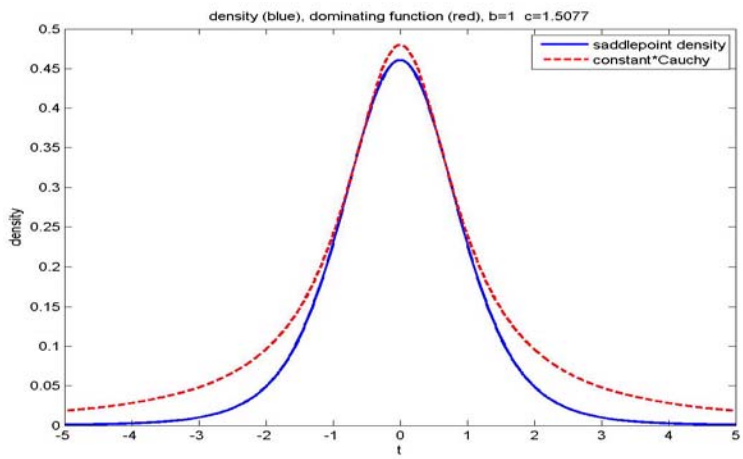

Figure 4: Saddlepoint density and dominating multiple of Cauchy density

"accept" T, i.e. output $X = k'(T)$ where

$$k\prime(T) = n\left(\frac{1}{\tanh T} - \frac{1}{T}\right)$$

- otherwise return to step 1.

The saddlepoint density and the dominating Cauchy multiple are shown in figure (4).

The output values are plot as a frequency histogram in Figure (5). The sample mean and variance of these 691446 values are 0.0017 and 1.3571 compared with their theoretical values $0, \frac{4}{3}$.

What is the price paid here for the use of an approximate distribution in place of the true distribution? In the center of the distribution the fit is nearly perfect, as one might expect. Indeed in Figure (6) the two cdfs are virtually identical, with a maximum difference of the order of 0.001.

Nevertheless the relative error of the approximation is larger in the extreme tails of the distribution. Figure (7) shows that the saddlepoint approximation is provides too little weight in the tail as we pass the extremes ± 3 and then piles up values very close to ± 4.This is a consequence of the fact that the derivative of the cumulant generating function $k'(t)$, although strictly increasing, has slope approaching 0 as $t \to \pm\infty$ in which case $k'(t) \to \pm 4$. The saddlepoint approximation is well-known to be excellent in the middle of a distribution, nearly perfectly in the interval (-3,3), but may fit less well in the extreme tails.

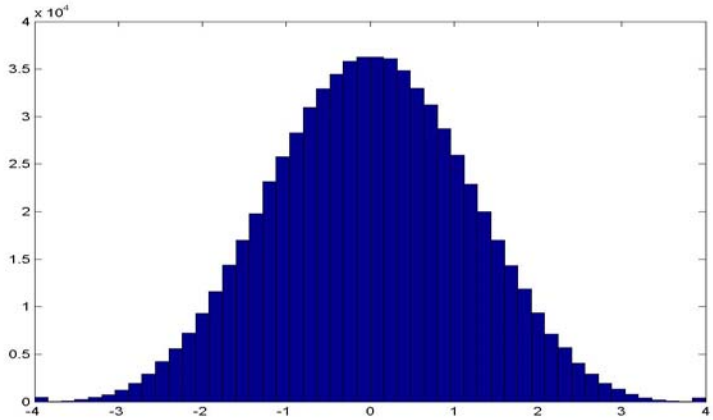

Figure 5: Frequency histogram of 691446 values obtained from saddlepoint approximation to distribution of sum of 4 uniform.

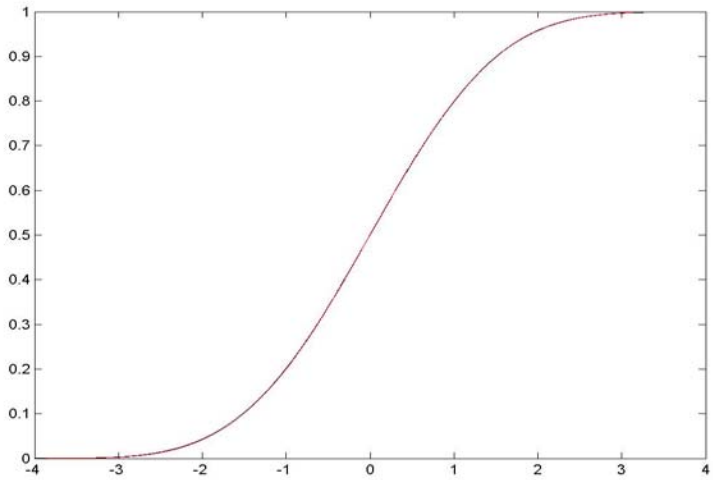

Figure 6: Empirical cdf of sample of 691561 from distribution of sum of 4 uniforms together with the saddlepoint approximation

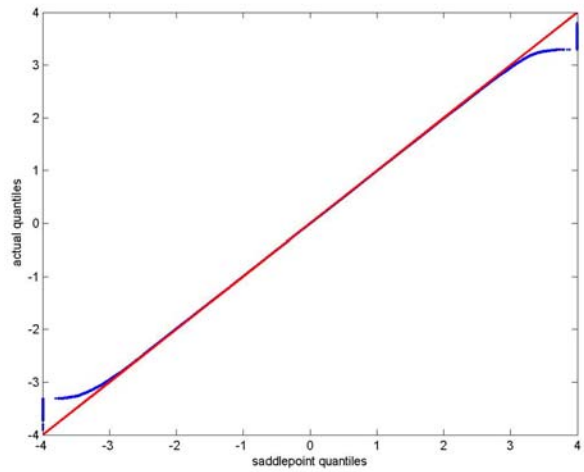

Figure 7: Quantile-Quantile plot, saddlepoint approximation vs true distribution for sum of 4 Uniform variates

As one might expect for the sum of 3 or fewer uniform, the fit of the normal base saddlepoint distribution is not nearly as good, because the saddlepoint approximation is an asymptotic approximation dependent in part on the central limit theorem. For other distributions, it may well be beneficial to us a base distribution other than the normal (see Wood, Booth and Butler, (1993)).

3 Conclusion

We have shown that simulation from a saddlepoint approximation is computationally and analytically efficient and, when only the transform of a distribution is available, is a reasonable alternative to the computationally intensive inversion of the moment generating function of a distribution. By using acceptance-rejection and the saddlepoint approximation, we are able to avoid the compuational expensive of inverting a Laplace transform or repeatedly numerical root-finding and still achieve a very good fit in a variety of examples.

References

[1] Barndorff-Neilson, O.E. and Cox, D.R. (1989) *Asymptotic Techniques for Use in Statistics.* Chapman and Hall, London

[2] Carr, P. and Madan, D., (2009) Saddlepoint Methods for Option Pricing. *The Journal of Computational Finance* 13. 49–61

[3] Devroye, L. (1986) *Non-Uniform Random Variate Generation*, Springer, New York

[4] Devroye, L., (2002) Simulating Bessel random variables, *Statistics and Probability Letters*, 57, 249-257

[5] Glasserman, P., Kim, K.K., (2009) Gamma expansion of the Heston stochastic volatility, *Finance and Stochastics*, forthcoming.

[6] Glasserman, P. (2004) *Monte Carlo Methods in Financial Engineering*, Springer, New York

[7] Heston, S.L., (1993) A closed-form solution for options with stochastic volatility with application to bond and currency options, *Review of Financial Studies*, 6, 327-343, 1993.

[8] Lugannini, R., Rice, S., (1980) Saddlepoint approximations for the distribution of the sum of independent random variables, *Advances in Applied Probability*, 12, 475-490

[9] Murakami, H. (2009) Saddlepoint Approximations to the Limiting Distribution of the Modified Anderson-Darling Test Statistic, *Communications in Statistics - Simulation and Computation*, 38: 10, 2214-2219

[10] Wood, T. A., Booth, J.G. and Butler, R.A. (1993) Saddlepoint Approximations to the CDF of Some Statistics with Nonnormal Limit Distributions, *Journal of the American Statistical Association*, 88, 680-686

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